

46. If $29 \times 48 = 576$, $35 \times 16 = 90$, $22 \times 46 = 96$, then $42 \times 17 = ?$

- a. 56
b. 286
c. 48
d. 64

47. If $235 = 38$ and $452 = 45$, then $345 = ?$

- a. 49
b. 66
c. 72
d. 50

48. $5 \times 6 \times 3 = 356$, $1 \times 0 \times 5 = 510$, $5 \times 6 \times 7 = ?$

- a. 567
b. 657

ANSWER KEY:

1. b	2. c	3. b	4. b	5. c
6. c	7. b	8. a	9. c	10. b
11. d	12. b	13. b	14. b	15. b
16. a	17. a	18. b	19. b	20. d
21. d	22. b	23. d	24. b	25. d
26. a	27. c	28. c	29. a	30. c
31. c	32. b	33. b	34. a	35. b
36. b	37. c	38. d	39. d	40. c
41. a	42. c	43. a	44. b	45. a
46. a	47. d	48. d	49. a	50. a

SOLUTION:

1. b

$7 \times 4 = 28$; $19 + 9 = 28$
 $8 \times 5 = 40$; $28 + 12 = 40$
 $9 \times 3 = 27$; $27 - 6 = 21$

2. c)

First figure : $6 \times 6 = 4 \times 9$
Second figure : $9 \times 8 = 24 \times 3$
Third figure $15 \times 6 = 9 \times ?$
Solving the above eqn we get
 $? = 10$

3. b)

Applying Pythagoras theorem in the third figure we get $? = 8$

- c. 210
d. 756

49. If $52 + 26 + 38 = 46$ and $24 + 36 + 52 = 34$, then $79 + 55 + 28 = ?$

- a. 104
b. 200
c. 98
d. 176

50. If $3454 = 5$ and $2332 = 5$, then $5245 = ?$

- a. 5
b. 10
c. 7
d. 8

4. b)

$(27 + 18) - (12 + 13) = 12$
 $45 - 25 = 20$;
 $(16 + 12) - (6 + 9) = 13$;
 $28 - 15 = 13$;
 $(10 + 11) - (5 + 4) = 12$
 $21 - 9 = 12$

5. c)

In each figure the lower left number is the square of the upper right number.
Again, First figure $3 + 9 = 7 + 5$
Second figure $2 + 8 = 4 + 6$
Third figure $4 + 7 = 5 + ?$
 $? = 11 - 5 = 6$

6. c)
The sum of cross products of the numbers gives the central number.

First Figure $(4 \times 6) + (6 \times 7) = 24 + 42 = 66$
Second Figure $(8 \times 9) + (5 \times 14) = 72 + 70 = 142$

Third Figure $(11 \times 7) + (9 \times 6) = 77 + 54 = 131$

7. b)

First Figure $(3)^3 + (2)^3 = 27 + 8 = 35$
Second Figure $(1)^3 + (5)^3 = 1 + 125 = 126$
Third Figure $(3)^3 + (4)^3 = 27 + 64 = 91$

8. a)

The sum of upper two numbers gives the lower right number while their product is equal to the lower left number.
First arrangement $5 + 4 = 9$ and $5 \times 4 = 20$
Second arrangement $3 + 8 = 11$ and $3 \times 8 = 24$
Third arrangement $9 + 4 = 13$ and $9 \times 4 = 36$

9. c)

First Figure $3 + 5 + 2 + 4 = 14$
 $14 + 13 = 27$
Second Figure $6 + 2 + 3 + 5 = 16$
 $16 + 21 = 37$
Third Figure $2 + 2 + 9 + 5 = 18$
 $18 + 29 = 47$

10. b)

Figure (1) $2 \times 3 \times 5 = 30$
Figure (2) $5 \times 1 \times (-1) = -5$
Figure (3) $4 \times 3 \times (-1) = -12$

11. d)

First Figure $4 \times 3 \times 2 = 24$
Second Figure $(-2) \times 2 \times (-1) = 4$
Third Figure $0 \times 6 \times 5 = 0$

12. b)

First Figure $(6 \times 7) + (8 + 4) = 42 + 12 = 54$
Second Figure $(8 \times 4) + (12 + 7) = 32 + 19 = 51$

Third Figure $(9 \times 5) + (14 + 9) = 45 + 23 = 68$

13. b)

First figure has $1^2, 3^2, 5^2$
Second figure has $2^2, 4^2, 6^2$
Third figure has $3^2, 5^2, 7^2$

14. b)

First figure $15 + 12 = 27$ and, $27 \div 9 = 3$

Second figure $44 + 28 = 72$ and, $72 \div 9 = 8$
Similarly, in third figure $64 + 53 = 117$ and, $117 \div 9 = 13$

15. b)

First of all take the product of all the numbers given outside the square and divide it by 10 to get the number inside the square.

First Figure $5 \times 3 \times 4 \times 2 = 120$

$120 \div 10 = 12$

Second Figure $5 \times 6 \times 2 \times 3 = 180$

$180 \div 10 = 18$

Third Figure $5 \times 2 \times 2 \times 9 = 180$

$180 \div 10 = 18$

16. a)

First figure $15 + 16 = 22 + 9$ i.e. $31 = 31$
Second figure $13 + 7 = 11 + 9$ i.e. $20 = 20$
Third figure $21 + 15 = ? + 13$
 $? = 36 - 13 = 23$

17. a)

First arrangement $(56 + 15) - (22 + 8) = 41$
 $71 - 30 = 41$
Second arrangement $(46 + 9) - (10 + 6) = 39$
 $55 - 16 = 39$
Third arrangement $(34 + 11) - (14 + 6) = 25$
 $45 - 20 = 25$

18. b)

$5 \times 6 + 3 \times 3 = 39$ and, $7 \times 5 + 4 \times 4 = 51$
Therefore $5 \times 5 + 3 \times 4 = 37$

19. b)

$(7 + 9 + 5 + 4) \times 2 - 10 = 40$
 $(17 + 8 + 3 + 6) \times 2 - 14 = 54$
 $(10 + 21 + 6 + 3) \times 2 - 18 = 62$

20. d)

First figure $3 \times 19 = 57$ and $57 + 1 = 58$
Second figure $81 \times 7 + 1 = 568$
Similarly, Third figure $93 \times 1 + 7 = 100$

21. d)

$(6 \times 4) + (10 \times 8) = 24 + 80 = 104$
 $(7 \times 8) + (9 \times 11) = 56 + 99 = 155$
Similarly, $(8 \times 9) + (5 \times 14) = 72 + 70 = 142$

22. b)

$4 + 2 + 3 + 7 = 16$ and $1 + 6 = 7$ $1 + 7 = 8$ and $8 - 7 = 1$
 $1 + 1 + 8 + 4 = 14$ and $1 + 4 = 5$ $1 + 3 = 4$ and $5 - 4 = 1$

$$7 + 4 + 9 + 3 = 23 \text{ and } 2 + 3 = 5 \quad 4 + 2 = 6 \text{ and } 6 - 5 = 1$$

23. d)

$$(0)2 + (3)2 + (1)2 + (2)2 = 0 + 9 + 1 + 4 = 14$$

$$(3)2 + (5)2 + (2)2 + (4)2 = 9 + 25 + 4 + 16 = 54$$

$$(3)2 + (1)2 + (6)2 + (5)2 = 9 + 1 + 36 + 25 = 71$$

24. b)

$$5 \times 3 + 1 = 16$$

$$16 \times 3 + 1 = 49$$

$$9 \times 3 + 2 = 29$$

$$29 \times 3 + 2 = 89$$

Therefore, $15 \times 3 + 3 = 48$

$$48 \times 3 + 3 = 147$$

25. d)

First Figure $14 + 12 + 7 + 3 = 36$

Second Figure $9 + 11 + ? + 16 = 54$

$$? = 54 - 36 = 18$$

26. a)

First Figure $(11 \times 12) - (6 \times 9)$

$$132 - 54 = 78$$

Second Figure $(14 \times 10) - (7 \times 8)$

$$140 - 56 = 84$$

27. c)

First arrangement $6 + 4 + 3 + 1 + 5 + 0 = 19$

$$19 \times 7 = 133$$

Second arrangement $2 + 5 + 3 + 4 + 6 + 8 = 28$

$$28 \times 7 = 196$$

Third arrangement $1 + 5 + 7 + 3 + 4 + 2 = 22$

$$22 \times 7 = 154$$

28. c)

First Figure $(4 \times 2) + (7 \times 1) + 3 = 8 + 7 + 3 = 18$

$$18 \times 8 = 144$$

$$1 + 8 = 9$$

Second Figure $(3 \times 3) + (5 \times 2) + 4 = 9 + 10 + 4 = 23$

$$23 \times 3 = 69$$

Third Figure $(6 \times 9) + (2 \times 3) + 4 = 54 + 6 + 4 = 64$

$$64 \times 4 = 256$$

$$6 + 4 = 10$$

29. a)

First Figure $5 \times 4 = 20 (= 10 \times 2)$

$$6 \times 2 = 12$$

Second Figure $6 \times 5 = 30 (= 10 \times 3)$

$$7 \times 3 = 21$$

Third Figure $4 \times 10 = 40 (= 10 \times 4)$

$$8 \times 4 = 32$$

30. c)

First Figure $(20 - 9)^2 = 121$

$$(11)^2 = 121$$

Second Figure $(24 - 11)^2 = 169$

$$(13)^2 = 169$$

Third Figure $(32 - 17)^2$

$$(15)^2 = 225$$

31. c)

The numbers in the given arrangement follow the rule given below :

$$4 \times 1 = 4$$

$$4 \times 2 = 8$$

$$8 \times 3 = 24$$

$$24 \times 4 = 96$$

$$96 \times 5 = 480$$

$$480 \times 6 = 2880$$

$$2880 \times 7 = 20160$$

32. b)

First column $9 \times 6 \times 5 = 270$

Second column $8 \times 7 \times 4 = 224$

Third column $8 \times 7 \times ? = 336$

$$? = 336 \div 56 = 6$$

33. b)

First column $9 - 1 = 8$, $8 - 1 = 7$

Second column $4 + 1 = 5$; $5 + 1 = 6$

Third column $2 + 0 = 2$; $2 + 1 = 3$ ($1 + 2$); $3 + 1 = 4$

34. a)

Proceed column wise. In each column the Second Number is equal to First Number + 1 and the Third Number is equal to First Number + 2 (or Second Number + 1).

1st Column $0 + 1 = 1$ and $1 + 1 = 2$

2nd Column $-1 + 1 = 0$ and $0 + 1 = 1$

3rd Column $-2 + 1 = -1$ and $-1 + 1 = 0$

35. b)

Proceed clockwise starting with the lowest number in the following manner :

$$7 \times 2 + 1 = 15$$

$$15 \times 2 + 1 = 31$$

$$31 \times 2 + 1 = 63$$

$$63 \times 2 + 1 = 127$$

$$127 \times 2 + 1 = 255$$

36. b)

$$5 + 20 + 6 + 9 = 40$$

$$4 + 8 + 15 + 3 = 30$$

$$9 + 25 + 7 + 9 = 50$$

$$22 + 7 + 8 + 8 = 45$$

37. c)

$$4 + 9 + 17 + 6 = 36$$

$$20 + 5 + 8 + 9 = 42$$

$$7 + 23 + 9 + 9 = 48$$

$$9 + 4 + 19 + ? = 40$$

$$? = 40 - 32 = 8$$

38. d)

$$4 - 3 = 1, \quad 1 - 1 = 0$$

$$9 - 6 = 3, \quad 6 - 2 = 4$$

$$12 - 8 = 4, \quad 6 - 3 = 3$$

$$12 - 10 = 2, \quad 4 - 1 = 3$$

39. d)

Different quadrants in question:

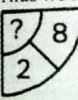


$$= 5 - 5 = 0 \rightarrow 0^2 \times 4 = 0$$



$$= 7 - 3 = 4 \rightarrow 4^2 \times 4 = 64$$

Thus we have,



$$= 8 - 2 = 6 \rightarrow ? = 6^2 \times 4 = 144$$

40. c)

First Column $= 4 \times 7 - 1 = 28 - 1 = 27$

Second Column $= 5 \times 8 - 2 = 40 - 2 = 38$

Third Column $= 6 \times 9 - 3 = 54 - 3 = 51$

41. a)

First arrangement $18 - 6 + 17 + 9$

$$12 + 17 + 9 = 38$$

Second arrangement $11 - 9 + 11 + 19$

$$2 + 11 + 19 = 32$$

Third arrangement $6 - 3 + 15 + 26$

$$3 + 15 + 26 = 44$$

Fourth arrangement $12 - 20 + 8 + 9$

$$29 - 20 = 9$$

42. c)

$$2 + 3 + 5 = 30$$

$$2 \times 3 \times 5 = 30$$

$$3 + 4 + 6 = 72$$

$$3 \times 4 \times 6 = 72$$

$$5 + 6 + 2 = 60$$

$$5 \times 6 \times 2 = 60$$

$$5 + 4 + 0 = 0$$

$$5 \times 4 \times 0 = 0$$

43. a)

$$20 \times 5 = 4$$

$$20 \div 5 = 4$$

$$48 \times 12 = 4$$

$$48 \div 12 = 4$$

Therefore, 77×11

$$77 \div 11 = 7$$

44. b)

$$72 - 9 = 63$$

$$37 - 9 = 28$$

Therefore $72 + 37$ is written as 6328

$$54 - 9 = 45$$

$$13 - 9 = 04$$

Therefore $54 + 13$ is written as 4504

Similarly, $61 - 9 = 52$

$$53 - 9 = 44$$

$61 + 53$ is written as 5244

45. a)

$$55 + 66 \rightarrow 5 + 6 = 11$$

$$11 \times 3 = 33$$

$$22 + 99 \rightarrow 2 + 9 = 11$$

$$11 \times 3 = 33$$

Similarly, $44 + 88 \rightarrow 4 + 8 = 12$

$$12 \times 3 = 36$$

46. a)

$$29 \times 48$$

$$2 \times 9 \times 4 \times 8 = 576$$

$$35 \times 16$$

$$3 \times 5 \times 1 \times 6 = 90$$

$$22 \times 46$$

$$2 \times 2 \times 4 \times 6 = 96$$

Therefore, 42×17

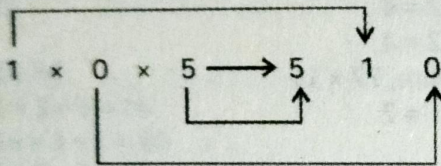
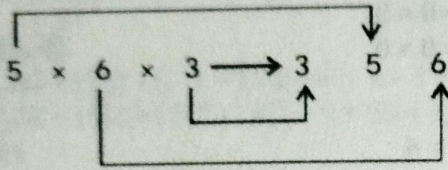
$$4 \times 2 \times 1 \times 7 = 56$$

47. d)

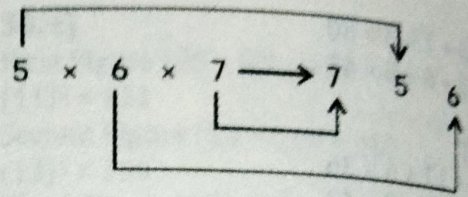
$$235 \text{ P } (2)2 + (3)2 + (5)2 = 38 \quad 452 \text{ P } (4)2 + (5)2 + (2)2 = 45 \quad 345 \text{ P } (3)2 + (4)2 + (5)2 = 50$$

48. d)

The solution is as shown below



Therefore,



49. a)

$$52 + 26 + 38 = 46$$

$$(5 \times 2) + (2 \times 6) + (3 \times 8) = 46$$

$$10 + 12 + 24 = 46 \quad 24 + 36 + 52 = 36$$

$$(2 \times 4) + (3 \times 6) + (5 \times 2) = 36$$

$$8 + 18 + 10 = 36 \quad 79 + 55 + 28$$

$$(7 \times 9) + (5 \times 5) + (2 \times 8)$$

$$63 + 25 + 16 = 104$$

50. a)

The required answer would be 5